MOHAMED AZIZ HADJAYED

Embedded Systems Engineer

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PROFILE

Final-year Embedded Systems Engineering student at the National Engineering School of Carthage. Passionate about Edge AI and IoT, with strong skills in C/C++, Python, TensorFlow/TFLite, and deploying lightweight AI models deploy on STM32 and Raspberry Pi platforms.

EDUCATION

National Engineering School of Carthage (ENICarthage), Ariana, Tunisia
Engineering Degree in Embedded Systems **Sep 2023 – 2026**

Preparatory Institute for Engineering Studies of Monastir, Monastir, Tunisia
Pre-Engineering Degree in Physics and Technologies, Rank: 90 / 630 **Sep 2021 – 2023**

PROFESSIONAL EXPERIENCE

E-commerce with Content Recommendation Engine, Mobelite, Monastir, Tunisia **15 July – 9 sep (2025)**

- Designed a web prototype using Streamlit, leveraging JSON files for user data storage and management.
- Implemented a content-based recommendation engine using scikit-learn, pandas, and NumPy.

Full-stack Developer Intern, DRÄXLMAIER(METS), Sousse, Tunisie **15 July - 30 August (2024)**

- Developed a full-stack web application using React, NestJS, and MySQL.
- Implemented a REST API for process management.
- Designed an intuitive user interface for workflow visualization.

PERSONAL PROJECTS

ESP32-Based Pass-Box Automation (Sterilization Cycle & Safety)

- ESP32 embedded system: 7-stage sterilization cycle, door interlocking, I2C LCD, MQTT/HiveMQ communication, FreeRTOS programming
- Complete IoT architecture: Real-time Node-RED dashboard, automatic CSV logging, dual email alert system (emergency + analytical reports), remote control

Embedded Autonomous Navigation System Based on TensorFlow

- Developed an autonomous robot on Raspberry Pi 4 using TensorFlow 2.x (CNN) for line detection.
- Implemented embedded vision, navigation AI, and motor control.

Security Evaluation Platform for Embedded Systems

- Developed a security evaluation solution on STM32F411RE.
- Built a monitoring interface using Flutter.
- Performed software and hardware attacks (UART, SWD/JTAG, Bleichenbacher).

IoT Solution for Automated Greenhouse Management

- Developed an ESP32-based system integrating multiple sensors for greenhouse monitoring.
- Designed an Android application for remote control and real-time data visualization.

Distributed IoT Network (Java)

- Sensor nodes + gateway + monitoring simulation using TCP sockets.
- Critical alerts via UDP multicast + distributed mutual exclusion (Lamport)

Sécurisation d'un système IoT (Raspberry Pi 4 – FastAPI + Flutter)

- 3 attaques sur Raspberry Pi4(serveur FastAPI) : DoS, reverse d'une app Flutter, extraction du firmware du SD.
- Mesure de l'impact (disponibilité/latence) et proposition de mitigations.
- Identification des faiblesses (configs/secrets/endpoints) et recommandations de contre-mesures.

TECHNICAL SKILLS

- **Languages** : Python , C, C++
- **Embedded & IoT** : STM32 (HAL/CMSIS), ESP32 (ESP-IDF), Raspberry Pi, Embedded Linux, FreeRTOS, Yocto Project
- **Data & AI** : NumPy ,pandas ,scikit-learn ,TensorFlow/Lite
- **Protocols** : UART, USART, I2C, SPI, CAN, MQTT
- **Frameworks & Tools** : FastAPI, Flutter, PlatformIO, MATLAB, KiCad, LabVIEW, ModelSIM
- **Dev Environment** : VS Code, STM32CubeIDE, STM32Cube.AI, GitHub, Node-RED, Firebase, Kali Linux

LANGUAGES

Arabe :Native language | French : B2 Level | English : B1 Level